

VIJAYA LAXMI

Embedded Software Engineer | Safety-Critical Systems | QNX & Linux Specialist

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PROFESSIONAL SUMMARY

Embedded Software Engineer with 4+ years of experience in safety-critical railway systems development at **Medha Servo Drives Private Limited**. Strong expertise in QNX OS, Embedded Linux, real-time systems, bootloaders, communication protocols, and controller bring-up. Proven experience in SIL-4 compliant system development (EN-50128) and complete embedded product lifecycle including V&V, field trials, RDSO testing, and ISA activities. Adept at low-level debugging, hardware-software integration, and kernel-level development support.

CORE COMPETENCIES

- Embedded C / C++ Programming
 - QNX RTOS (Momentics IDE, BSP integration)
 - Embedded Linux & Kernel-Level Debugging
 - Safety-Critical Systems (SIL-4, EN-50128)
 - Bootloader Development (TI RM57)
 - Communication Protocols: CAN, UART, SPI, I2C, Ethernet, TCP/IP
 - DMA, Flash Memory, EEPROM, FRAM
 - GSM / GPS / Radio Modem Integration
 - Board Bring-Up & Hardware Debugging
 - Tools: CCS, MCUXpresso, Eclipse
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PROFESSIONAL EXPERIENCE

Embedded Software Engineer – R&D

Medha Servo Drives Private Limited | 4+ Years

- Designed and developed safety-critical embedded software for Train Collision Avoidance System (TCAS).
- Ported and optimized KAVACH software stack onto QNX OS hardware platform.
- Developed bootloader and low-level drivers for RM57 controller.
- Implemented DMA-based UART communication and high-reliability Ethernet interfaces.
- Integrated GSM module for secure KMS server communication over TCP/IP.
- Developed Event Logger Card for centralized railway data logging via Ethernet.

- Engineered Temporary Speed Restriction Management System (TSRMS), a fail-safe redundant subsystem.
 - Executed controller board bring-up and peripheral validation testing.
 - Developed software for Flash, EEPROM, FRAM, and serial interface memories.
 - Involved in full SDLC: Requirements → Design → Implementation → V&V → Field Trials → RDSO Functional Testing.
 - Supported cross-functional system validation and ISA compliance processes.
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KEY PROJECTS

TCAS – Moving Block Implementation

Enhanced collision avoidance system for Indian Railways with improved computational performance on QNX OS platform, increasing track capacity and operational safety.

RIU Simulator Development

Built simulator to emulate field inputs and communicate with Vital Computer Card for validation and system-level testing.

Temporary Speed Restriction Management System (TSRMS)

Designed automatic speed enforcement subsystem for emergency and maintenance railway scenarios.

Event Logger Card

Developed Ethernet-based logging system with GSM-based secure session key establishment for TCAS infrastructure.

TECHNICAL ENVIRONMENT

- **Operating Systems:** QNX RTOS, Embedded Linux
 - **Languages:** C, C++, Python
 - **Microcontrollers:** TI RM57, NXP, ARM Cortex series
 - **Debugging:** JTAG, CCS, MCUXpresso, Momentics
 - **Standards:** EN-50128 (SIL-4), Railway Safety Compliance
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EDUCATION

B.Tech – 8.2 CGPA

Rajiv Gandhi University of Knowledge Technologies

CERTIFICATIONS

- Machine Learning – NPTEL
- Deep Learning – Amity Future Academy